

Computer-Vision-Guided Human Pulse Rate Estimation: A Review

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Abstract—Human pulse rate (PR) can be estimated in several ways, including measurement instruments that directly count the PR through contact- and noncontact-based approaches. Over the last decade, computer-vision-assisted noncontact-based PR estimation has evolved significantly. Such techniques can be adopted for clinical purposes to mitigate some of the limitations of contact-based techniques. However, existing vision-guided noncontact-based techniques have not been benchmarked with respect to a challenging dataset. In view of this, we present a systematic review of such techniques implemented over a uniform computing platform. We have simultaneously recorded the PR and video of 14 volunteers. Five sets of data have been recorded for every volunteer using five different experimental conditions by varying the distance from the camera and illumination condition. Pros and cons of the existing noncontact image- and video-based PR techniques have been discussed with respect to our dataset. Experimental evaluation suggests that image- or video-based PR estimation can be highly effective for nonclinical purposes, and some of these approaches are very promising toward developing clinical applications. The present review is the first in this field of contactless vision-guided PR estimation research.

Index Terms—Color-based analysis, face tracking, linear decomposition, motion-based analysis, noninvasive method, nonlinear decomposition, pulse rate (PR) estimation.

I. INTRODUCTION

HUMAN pulse rate (PR) is one of the vital parameters that can indicate physiological state of living persons. Therefore, knowledge about pulse is a prerequisite to understand health condition of a person. Since PR is directly associated with heartbeat, more often than not, physicians initially check PR before undergoing minute analysis of the heart functionally through complex procedures such as electrocardiography (ECG).

Normal PR for healthy adults varies between 60 and 100 beats/min [1]. However, females aged 12 or above tend to have faster PRs than the males and PR of infants below four years of age may vary between 80 and 120 beats/min [2]. Human heart can beat too fast or too slow due to several factors. For examples, it may vary and increase with exercise, illness, injury, and emotions [3]–[6]. PR usually shoots up in response to adrenaline,

preparing the body for a quick reaction. On the other hand, being frightened or surprised can automatically make the heart rate (HR) higher via adrenaline, preparing us to use more oxygen and energy for the reflex reaction. When the PR of a normal person at rest goes over 100 beats/min, the condition is known as tachycardia, whereas bradycardia refers to a condition when the heart is beating too slow—usually below 60 beats/min [7].

Abnormalities in rhythms of the heart are reflected through QRS complex and are often used for detecting blockages, tachycardia, and bradycardia [7], [8]. Physicians usually perform routine check-up by recording the PR of a patient before applying complex analysis. Thus, accurate estimation of PR is equally important. Body-contact-based measurement by checking vibrations in arteries running beneath the skin at the wrist or neck is a gold standard approach. Other contact-based methods include ECG, phonocardiography (PCG), echocardiography (Echo), thermal imaging, photoplethysmography (PPG), blood pressure wave measurement, optical scattering phenomenon, Doppler phenomenon, measurement using ultrasonic devices, and measurement with ultrawideband (UWB) radars. Contactless PR estimation techniques are also used. Such techniques have certain advantages over contact-based techniques. A recent review on noncontact-based techniques can be found in [9].

In recent years, several computer-vision-guided PR detection techniques have been proposed, e.g., [10]–[19]. Though the fundamental concepts behind these techniques are well established, however, effectiveness of these methodologies in real-life scenario depends on various factors such as variation in illumination, motion artifacts, distance from the camera, quality of the imaging sensors, etc. It has been felt that the existing vision-guided contactless methods need be studied in depth and analyzed rigorously before they can be adopted in clinical processes. To the best of our knowledge, such techniques have not been benchmarked yet. In this paper, we present a comprehensive review on existing contactless, vision-guided, and noninvasive PR estimation techniques and benchmark them with respect to accuracy, computational overhead, and clinical applicability.

A. Motivations and Contributions

Among all available methods of PR detection, video-based monitoring methods are the cheapest and largely suitable for nonclinical purposes [10]. Some of these methods try to extract PPG signal by analyzing the periodic changes in skin color and others take advantage of pixel tracking techniques to detect subtle periodic motion generated in the body and transmitted to head, caused by cardiac pumping action. The

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remote photoplethysmography (rPPG) is based on the principle that absorption of light by blood cells is more as compared to the surrounding tissues, hence any variation in blood flow essentially incorporated into the reflected light from the nearby areas. The motion-based technique analyzes a video to detect subtle motion change which are generally not observable through naked eyes [20]. PR obtained from PPG or rPPG is a low-cost noninvasive means of sensing cardiovascular blood volume pulse (BVP) variations through transmitted or reflected light [21]. However, to establish the acceptability of such methods, they need to be benchmarked against gold standard practices. This has motivated us to study various noninvasive, contactless, and vision-guided PR detection methods and benchmark these techniques. Following are the main contributions of this paper.

- 1) Implementation of the major state-of-the-art techniques of computer-vision-guided contactless PR estimation on a uniform platform.
- 2) Creation of a large dataset comprises with ground truth PR value and video recording of 14 volunteers. Data for each volunteer have been recorded using five different experimental conditions by varying the distance of the user from the camera and varying environmental illumination conditions.
- 3) A comparative analysis to benchmark state-of-the-art video and image-guided noninvasive PR detection techniques with respect to statistical parameters, such as mean absolute error, standard deviation (SD) error, mean square error rate, correlation coefficient, and computational overhead, has been performed.

Rest of the paper is organized as follows. In Section II, we present a comprehensive overview on evolution of noninvasive PR detection techniques. Next, we present a technical walk-through of vision-guided noninvasive PR detection technique. Experimental setup, datasets, and parameters related to various methods are discussed in Section IV. This is followed by experimental evaluations in Section V. Discussion of results is presented in Section VI. Finally, we conclude in Section VII.

II. EVOLUTION OF PR DETECTION TECHNIQUES

This section presents a review of the state-of-the-art methods. Research work carried out in this area varies from manual to automatic PR extraction, and they can be categorized into two classes: 1) invasive and 2) noninvasive. Both techniques find applications in HR monitoring depending on the state of a patient. Invasive techniques developed so far include Fick's method [22], [22]–[25], indicator dilution [26]–[28], and thermodilution [29]–[32]. Out of these three techniques, thermodilution is most often used in clinics for monitoring PR of surgical patients. However, the method has certain disadvantages. For example, it can only be conducted by highly skilled surgeons [33]. Thus, research is mostly shifting toward an easy and noninvasive means of PR measurement.

A. Noninvasive Techniques

PR can be measured using noninvasive way through ECG, PCG, Echo, PPG, thermal imaging, blood pressure wave

measurement, optical scattering phenomenon, Doppler phenomenon, ultrasonic devices, and UWB radars. These techniques can further be grouped into contact-based and contactless methods. Though ECG and PCG are the gold standard for PR measurement, yet they require experts to carry out the measurement. Also, these techniques are expensive, uncomfortable, and often unsuitable for continuous monitoring. Hence, contactless techniques are getting attention of various researchers.

UWB-based PR measurement has gained a lot of momentum in recent years due the feasibility measurement using wireless systems. It is based on the Doppler phenomenon [34]–[37]. The technique uses a Doppler radar for transmitting a continuous microwave signal and detecting the phase modulation in the received signal caused by the chest-wall displacement due to pulsating effect of the heartbeats. The basic components for UWB monitoring system involves transmitter, receiver, reflective target (the patient), and a signal-processing unit. The main advantage of using UWB radar for signal acquisition is its ability to measure the signal even when the source is behind an obstacle [38]–[40]. Therefore, the method can be applied in surveillance, security, rescue operations, and clinical purposes.

Recent research also reveals that carotid pulse can be measured using optical devices with the help of Doppler phenomenon such as laser Doppler vibrometry [41]–[44]. Light incident on a vibrating or moving object is scattered, making a slight modulation in phase and frequency. A high-precision laser device can measure the frequency modulation of the scattered light when superimposed with the reference beam incident on the output photo detector. This superimposed signal contains the information regarding the Doppler shift in the frequency. Efficient signal-processing techniques can revive the velocity caused by vibration and displacement of the test object by analyzing the superimposed signal. This principle is utilized in measuring the PR of a person by directing the laser beam on the superficial blood vessels of body parts like neck region overlying the carotid artery [45].

PR can also be measured from the blood flow waveform of carotid artery by a method known as Doppler Echo. Holdsworth *et al.* [46] have demonstrated the recovery of the pulse wave from carotid on the basis of blood velocity spectra. It is a noninvasive technique for quantifying cardiac output using Ultrasound [47]–[50] to examine the direction of blood flow in heart or blood vessels using the Doppler effect. The blood velocity through heart results in a Doppler shift in the frequency of the reflected ultrasound waves. This Doppler shift can be used to effectively calculate cardiac output from the flow velocity and volume of the 2-D ultrasound imaging waveform. The periodic pattern in the flow of blood is then used to determine the HR. Doppler ultrasound technique is commonly used in the estimation of fetal HR and HR variability [51]–[53].

PR can be measured by analyzing the thermal image of major superficial vessels using highly sensitive thermal imaging system as demonstrated by Garbey *et al.* [54]. Modulation of temperature is reflected in thermal signals obtained from the vessel due to the pulsation effect of blood flow. Next, a fast Fourier transformation (FFT) is applied on line-based region of the vessel to extract pulse frequency.

Piezoelectric transducers or sensors can be used to measure pulse frequency based on indirect mechanical effects generated by the arterial pressure change due to the blood flow. A piezoelectric device [55] outputs voltage signal corresponding to the mechanical stress subjected to it. Local blood pressure measurement reflecting the HR of a subject can be obtained from the fingertip and a corresponding voltage waveform can be drawn. Despite high reliability of the technique in comparison with other noninvasive methods, it suffers from motion artifacts due to subject movement. Recently, Kharrat *et al.* [56] have developed an adaptive PR measurement technique through piezofilm sensor using ensemble empirical mode decomposition from noisy data with motion artifacts.

Several other methods that use fiber-optic sensors have also been introduced to detect pulse frequency [57], [58]. Measurements using fiber optics involve a light-sensitive device that detects any perturbations in the physical length of the optical fiber due to external disturbances. Very recently, Yang *et al.* [59] have reported that a textile fiber-optic microbend sensor can be used for heartbeat and respiration monitoring. This type of optical sensors can be embedded in a belt [59] or behind a pillow or bed [60]–[62] to measure vital signals generated as vibration of the pulse pressure.

Plethysmography [63] is a method of measuring volumetric change in different parts of the body resulting from fluctuation in the amount of blood flowing through arteries and veins. Signal acquisition can be done in several ways [64]–[70]. It can be used for detection of HR since the change in blood volume is related to PR. Plethysmography when associated with optical technique is known as PPG. It can be done in two ways, namely transmittance [71] and reflectance [72], depending on the placement of the light source and detector. Pulsoximetry [73] uses the basic principle of PPG for monitoring a person's O_2 saturation level. Oxygenated blood absorbs more infrared light (850–1000 nm) and deoxygenated blood absorbs more red light (600–750 nm wavelength). Hence, HR can be measured from the periodic pulse pressure wave generated during regular circulation of oxygenated and deoxygenated blood. Previously, PPG was sensed from feet, fingertip, and chest; however, researchers have shown that the signal can be obtained from facial image and video [10]–[18].

B. PR Using Smartphones

Advancement in the smartphone technique has open up a new way of estimating various physiological parameters including PR. These devices are often equipped with highly accurate sensors such as accelerometer, gyroscope, magnetometer, touch screen, high-definition camera, proximity sensor, etc. Therefore, researchers have tried to estimate various physiological parameters such as PR, body temperature, BP, breathing rate, cardiac R–R intervals, or blood oxygen saturation using data collected from these sensors attached with handheld devices. For example, Scully *et al.* [74] have proposed a smartphone-based methodology to record breathing rate, cardiac R–R intervals, and blood oxygen saturation. PR estimation using smartphone has been attempted by other research groups for different

purposes including atrial fibrillation (AF) detection. Existing methods include techniques proposed by Choudhury *et al.* [75], Pal *et al.* [76], Lee *et al.* [77], and McManus *et al.* [78].

The authors of [75] have used a two-stage approach for robust HR calculation from the PPG signal captured using smartphones. First, normalized least mean square (NLMS)-based adaptive filter has been used to clean up the noisy PPG signal which is followed by HR calculation from the frequency spectrum. The results are further fine-tuned using different interpolation techniques. The method proposed by Pal *et al.* [76] uses a two-stage approach. First, the onset of good video signal is detected by formulating a finite-state machine that employs multiple-window short-time FFT. Only upon receiving sufficient acceptable video signal, the HR is computed.

Similarly, McManus *et al.* [78] have obtained pulsatile time series recordings before and after cardioversion using an iPhone 4S camera. They have used a novel smartphone application for real-time pulse analysis using two statistical methods: root mean square of successive RR difference (RMSSD/mean) and Shannon entropy (ShE). They have examined the sensitivity, specificity, and predictive accuracy of both algorithms using the 12-lead electrocardiogram as the gold standard. AF detection can be done by recording pulsatile photoplethysmogram signal from a fingertip using the built-in camera lens [77].

III. COMPUTER-VISION-GUIDED PR ESTIMATION

According to the different camera-based techniques, PR estimation can be broadly categorized into two classes.

- 1) *Image-guided techniques*: They mostly use the intensity levels of different color channels over time to build the feature trajectory to be fed into some statistical model for PR estimation.
- 2) *Motion-guided techniques*: They make use of tracking key points over time to estimate the feature for pulse estimation.

In this section, we have summarized and compared the existing vision-guided PR estimation techniques as mentioned in Table I. These techniques have been summarized with respect to advantages and disadvantages. Next, we have discussed some of the key statistical methods and frequency-domain analysis that are pivot to these techniques.

A. PR Estimation Through Image Analysis

Image-analysis-driven techniques are based on the concept of rPPG [82] for measuring vital signs from a human body. The blood volume all over the body changes with the rhythmic pulsating action of heart. This results in generation of cardiac pulse which is carried toward the peripheral tissues of the body like palm, fingertip, and skin. Therefore, the light reflected from a skin region also undergoes a periodic change in the amount of light reflected due to the blood volumetric change. In the past few years, several researchers have exploited the above fact to extract the vital signs from human face image captured using an ordinary commercial camera [10]–[14], [17], [19], [80]–[84] as well as advanced camera [85].

TABLE I
SUMMARY OF IMAGE- AND VIDEO-GUIDED PR ESTIMATION TECHNIQUES

Algorithm	Year of publication	No. of subjects	Frame rate (frames/s)	Feature type	Image/Video	Statistical method	Estimation technique	ROI	ρ /Accuracy (RMSE in beats/min)	Own Database	MAHANOB-HCI Database [79]
Takano and Ohta [19]	2007	14 (F)	5	Intensity	Image	AR spectral analysis	B-spline Interpolation	cheek	0.9 (ρ)	-	
Poh <i>et al.</i> [10]	2010	12 (10M, 2F)	15	RGB	Image	ICA	FFT	Face	2.29	25.9 [17]	
							ZCR	Face	2.51		
Lewandowska <i>et al.</i> [13]	2011	10 (8M, 2F)	20	RGB	Image	PCA	ZCR	Forehead	11.37	-	
							FFT	Face/Forehead	1.96		
Poh <i>et al.</i> [11]	2011	12 (8M, 4F)	15	RGB	Image	ICA	LP	Face	1.24	13.6 [17]	
						LE			0.02		
Wei <i>et al.</i> [14]	2013	20 (11M, 9F)	30	RGB	Image	LE	IBI	Face	0.02	-	
						Isomap			9.99		
						LLE			0.58		
						LTSA			0.52		
Balakrishnan <i>et al.</i> [15]	2013	19 (11M, 7F)	30	key-points	Video	PCA	FFT	Face	1.18	21 [17]	
Irani <i>et al.</i> [16]	2014	5	30	key-points	Video	PCA	DCT	Face	0.93	-	
Li <i>et al.</i> [17]	2014	10 (8M, 2F)	30	G channel	Image	NLMS	WP	Face	1.27	7.62 [17]	
Monkarese <i>et al.</i> [80]	2014	10 (8M, 2F)	30	RGB	Image	ICA	kNN	Face	3.64	-	
							Regression		7.31		
Kwon <i>et al.</i> [12]	2012	10 (8M, 2F)	29.99	G channel	Image	-	FFT	Face	1.46	25.1 [17]	
				RGB		ICA			6.19		
Yu <i>et al.</i> [81]	2013	Not reported	25	RGB	Image	ICA	STFT	Face	2.19	-	
Tasli <i>et al.</i> [82]	2014	10	30	G channel	Image	RPPG method	inverse FFT	Face	5.93	-	

Note: ROI—Region of interest; NLMS—Normalized least mean square; LP—Lomb periodogram; IBI—interbeat interval; WP—Welch's periodogram; kNN—k-nearest neighbor; STFT—Short-time Fourier transform; RPPG—Remote photoplethysmography; ρ : Correlation coefficient.

One of the earliest studies published in this direction is [19]. They tracked the average brightness of the manually selected region of interest (ROI) on the subject's face. The signal is processed, filtered and an autoregressive (AR) spectral analysis is employed to find the dominant frequency corresponding to PR. However, this method estimates the average PR over a 30-s period and the numerical processing is simply based on a trial-and-error testing. They failed to give any concrete mathematical reasoning for adopting such a method. A similar work has been proposed in [86] that estimates PR by spatially averaging R, G, and B color channels from a manually selected ROI. They have used FFT for pulse frequency estimation after filtering through a bandpass Butterworth filter in a phase neutral (forward and reverse) digital filter. However, both methods have defined a fixed ROI for signal acquisition. Therefore, the observed region must be kept stable and even minor movements are not permitted. Moreover, they have not compared how different channels give PR estimation even after agreeing with the argument given in [70] that green channel contains the strongest phelthysmographic signal.

1) Measurement Using Poh's Method: The work that have gained high popularity in this field of research is Poh's work [10]. Poh *et al.* have explored the possibility of measuring PR from a web-cam. They made the ROI selection automated using Viola-Jones [87] face detection algorithm and computed the mean intensity values of the detected ROI for every frame for each of the R,G,B color channels. They have employed independent component analysis (ICA) to separate PPG signal from the three color traces. They have typically selected the second ICA component as their desired PPG signal and transferred into frequency domain using FFT. The frequency

corresponding to the highest power in the operational range of [0.75, 4] Hz corresponds to the PR frequency. Poh's results show that ICA separated sources can achieve a higher accuracy as compared to green channel trace.

Pros and Cons: They are the first who have used higher order statistics for an effective component separation based on joint approximate diagonalization of eigenmatrices (JADE) algorithm, which is unique in nature. However, the method suffers from certain flaws in terms of accuracy as it is based on the linearity assumption for component extraction from three channels which may not be true always. Also, neither there is any pre-processing stage to handle environmental noise nor any mention regarding the distance between face and camera. Although, Poh *et al.* [11] have improved their method by adding several temporal filters before and after applying ICA, however, scaling issue still remains as a persistent problem.

2) Measurement Using Lewandoska's Method: Lewandoska *et al.* [13] in turn showed that it is possible to extract PR from a small rectangular region of the face (i.e., forehead) using only two color channels. They have used principal component analysis (PCA) instead of ICA for component separation and claimed that cardiac pulse extracted using PCA is comparable to that obtained by ICA. Thus, there is no need for using computationally expensive step such as ICA.

Pros and Cons: The method is less time consuming since PCA takes less time as compared to ICA. The computational time using ICA is approximately 223 ms, while in the case of PCA, it is only 1.4 ms, as reported in [13]. Moreover, uniformity of ROIs selected from the forehead region and absence of any moving element increases the accuracy of measurement. On the other hand, decrease in the ROI size (in case of

forehead) increases the noise level as compared to face region. It also imposes a constraint as the subject needs to be motionless and must be in sitting or supine position. Illumination is also important since underexposure or overexposure may result in loss of information.

3) Measurement Using Wei's Method: Wei *et al.* [14] took a different approach from [13] for BVP extraction. According to the Beer–Lambert Law, light intensity traversing through facial tissues vary nonlinearly with distance. Hence, neither PCA nor ICA could extract pure BVP as both are based on linear hypothesis. The authors have carried out experiments with several nonlinear dimensionality reduction techniques such as Laplacian eigenmap (LE) [88], Isomap [89], local linear embedding (LLE) [90], local tangent space alignment (LTSA) [91], maximum variance unfolding [92], and linearity preserving projection [93]. They came to a conclusion that LE produces the best results.

Pros and Cons: The main reason behind choosing this method is that it can handle motion artifacts and variation in illumination condition in a better way as compared to linear reduction techniques. Furthermore, in ICA, component selection is merely a matter of the user's choice that can affect the results. The method is computationally expensive as compared to linear techniques, however, less complex than other nonlinear techniques.

4) Measurement Using Monkaresi's Method: Monkaresi *et al.* [80] have introduced a machine learning approach to improve contactless pulse monitoring using a webcam. According to them, the performance consideration of any single ICA component is not sufficient in PR estimation. By using the independent components as a cue, they have trained a power spectral density model for feature extraction. Then two machine learning approaches, namely linear regression and k-nearest neighbor (k-NN) classifier, were used to estimate the PR from the computed values. The root-mean-square error (RMSE) reduced from 43.76 to 3.64 beats/min as compared to Poh's [10] method when experimented on their own dataset.

Pros and Cons: Feasibility of remotely sensing physiological parameters in naturalistic human–computer interaction (HCI) scenarios and indoor exercising condition have never been tested in any of the previous methods. They have concluded that the k-NN approach outperforms other techniques since it tries to select the best independent component. Though accurate pulse measurement in presence of movement is quite encouraging, however, their method cannot handle illumination variation and fluctuation in environmental parameters.

5) Measurement Using Yu's Method: Yu *et al.* [81] have used short-time Fourier transform (STFT) instead of FFT unlike Poh *et al.* [10] for determining the PR. Typically, STFT provides more accurate localized temporal and frequency information. Thus, they have used the area between the eyes and upper lip of the mouth as the ROI for accurate reading. After performing ICA for component separation, the one with the highest power in the operable range was used as input data for STFT.

Pros and Cons: Their work mainly deals with the rapid change in the PR trend, like in exercising condition or cycling condition unlike other works that mainly deal with situations when the PR is in steady condition.

6) Measurement Using Lee's Method: Possibility of PR measurement by analyzing the color features from skin region have been demonstrated by Lee *et al.* [94]. This contact-free method uses a camera focused on the skin regions such as head/neck, inner arm, outer arm, palm area, etc. They have used ambient light to measure the PR. In this works, a neural network architecture, multilayer perceptron has been used as a skin classifier after training it with several manually labeled skin pixels. A weighted sum of three color channels has been used as a raw signal and discrete Fourier transform is used along with a novel data adjustment scheme.

Pros and Cons: The method can be simultaneously applied on multiple body parts for pulse estimation. There is a chance that the ROI pixels may be confused with skin pixels. Body movements and light effects are also major concerns.

7) Measurement Using Tasli's Method: Tasli *et al.* [82] used the active appearance model [95] technique to correctly detect and track facial landmark of a selected ROI for robust signal acquisition. Hence, the ROI is generated irrespective of the subject's expression and talking during the data acquisition stage. They have used the average green channel as the raw signal removing the outlier pixels while calculating mean for a selected ROI and applied conventional detrending technique [96] to extract the low pass equivalent of BVP signal.

Pros and Cons: The method may be effective in artifacts handling, yet a poor lighting condition may cause degradation of the performance.

8) Measurement Using Li's Method: One interesting work was done by Li *et al.* [17] to measure PR from face under realistic situation. Their main focus was illumination correction and nonrigid motion elimination. They have used discriminative response map fitting [97] to detect and generate a mask of ROI from the facial landmarks and then used Kanade–Lucas–Tomasi [98] algorithm to track the ROI from face. The average green value of the selected ROI was computed as raw signal. Illumination correction was made by background separation. Next, its average green value is taken as a reference to model illumination variation using the NLMS algorithm. The overall signal was segmented and they computed the SD for each segment. The segment with SD greater than a threshold was removed to eliminate the nonrigid motion followed by a series of temporal filtering to exclude the frequencies outside the range of interest.

Pros and Cons: The above method performs satisfactorily in steady conditions; however, it shows significant improvement when tested over MAHANOB-HCI public database as compared to other methods [10]–[12], [15]. Though their method can handle illumination variance and nonrigid movements, however, extreme rotation of the head, especially in yaw direction, may result in loss of feature points as half of the face are lost.

9) Measurement Using Wang's Method: The main drawback of Li's [17] method, i.e., discontinuous face localization, was effectively handled by Wang *et al.* [84] by a new strategy to track pixels of the interested regions. Here, they have treated each pixel as an independent PPG sensor instead of spatially averaging into a single trace unlike other methods. Global and local motion compensation was addressed using circularity structure of tracking-by-detection with kernels [99] and

forward-backward flow tracking [100] algorithms. The pixels with large abrupt motions are removed as noise. The spatial redundancy of the remaining pixels are thus disintegrated by spatial pruning technique in a temporally normalized color space to obtain multiple pulse traces. Finally, they have used bandpassed filters and combined the temporally extracted pulse-traces into a single optimized rPPG signal.

Pros and Cons: This method typically integrates the motion tracking and color-based PPG method in PR estimation as compared to other methods. They have proposed a novel algorithm to create and optimize pixel-based rPPG sensors in spatial and temporal domain for efficient pulse measurement. The signal-to-noise ratio was improved from 3.34 dB of the state-of-the-art work to 6.76 dB with some tolerance.

10) Measurement Using Haan's Method: The work of Haan and Jeanne [83] focuses on motion handling problems at the raw data acquisition stage. They benchmarked various chrominance-based algorithms that use the blind source separation (BBS) technique and compared them with their method for selection of proper signal. They tried to explore the possibility of eliminating the additive component of the raw R, G, and B traces that do not contribute to the PPG signal. A previous work [101] suggests that the diffused component of light that has traversed through skin undergoes the Cardiac pulse variations unlike the optical reflection component that carries no pulse. Therefore, they have investigated on the skin-tone standardization to find a fixed optimal weighing coefficients rather than using the heuristic-based approach proposed by Poh *et al.* [10] or Lewandowska *et al.* [13] for seeking the correct components.

Pros and Cons: The important advantage of their method is that it completely eliminates the heuristic selection of weak component and can quickly adapt to changing conditions with shorter latency. In terms of motion robustness, Haan's method performs better than traditional BBS techniques of PCA and ICA for separation of pulse signal from the periodic distortion.

11) Measurement Using Kwon's Method: The state-of-the-art smartphones are equipped with high-resolution cameras. Hence, the built-in camera system of smartphone can be used for PPG signals extraction. Scully *et al.* [102] and Jonathan and Leahy [103] tried to measure the PPG signal from the fingertip placed in contact with the optical sensor. However, Kwon *et al.* [12] used an alternate way of measuring PR from distant as a validation step of Poh's work [10]. The method adopted by Kwon *et al.* was similar to Poh's method except the measurement was done from a smartphone built-in camera.

Pros and Cons: The main advantage of using smartphone is that it can be used anywhere due to its portability. However, smartphone-based methods are prone to motion artifacts as the accuracy depends on the relative motion between user and phone.

B. Motion-Based Analysis

A person sitting in an upright posture undergoes a mixed motion due to the pumping action of the heart along with several other involuntary motion. This results in subtle head motion at each cardiac beat due to Newtonian reaction of blood influx.

Exploiting this idea, MIT developed a method [20] called as "Eulerian video magnification" to amplify these subtle changes that are normally not visible by naked eyes. Based on these video magnification, several works [15], [16], [104] have been published in recent years for extracting PR. This technique of pulse measurement also possesses certain advantages over the chrominance based methods. For example, it is not restricted to any particular view of the face. Another advantage is that image-based methods often require color image frames, whereas motion-based techniques can work on color as well as grey scale videos.

1) Measurement Using Balakrishnan's Method: Balakrishnan *et al.* [15] first suggested the PR extraction by tracking subtle head oscillations caused by cardiac pulsating action. They used PCA to segregate pulse signal from the trajectories of multiple feature points. Only the vertical component of the trajectory was used for computation since most of the oscillation was assumed to be on the vertical direction. The PCA decomposed signal was then filtered using Butterworth filter followed by FFT. The method achieved promising results on their self-acquired database; however, it did not find much applicability.

Pros and Cons: Since the method uses several key features, a choice can be made for less erratic feature points for accurate pulse extraction. As the method relies on motion tracking, subjects must avoid possible voluntary motions. The subjects must be in a sitting or supine position during measurement. Suboptimal lightning condition or the presence of noise may affect the results as the motion due to cardiac oscillation itself has a very low amplitude and may affect feature trajectory.

2) Measurement Using Irani's Method: The method of Balakrishnan *et al.* [15] was improved by Irani *et al.* [16] by replacing FFT with discrete cosine transform (DCT) and adding smoothing filter before the bandpass filter to handle motion artifacts. The state-of-the-art work of Balakrishnan was tested along with their modification over subjects with different facial actions like smiling, speaking, angry, and with changing head pose. They have found that their method performs better as compared to Balakrishnan's method.

Pros and Cons: The method is expected to give better performance since the dimension of features point trajectories is larger as compared to R,G,B (only three feature trajectories); however, at the same time, it is susceptible to motion distortion. Thus, this method gives accurate results under minor motion conditions, and it suffers from scaling issues, as a larger distance between camera and the subject reduces the performance due to tracking problems in the presence of noise.

3) Measurement Using Shan's Method: Shan's [104] work is no different from Balakrishnan *et al.* [15] and Irani *et al.* [16], except they have demonstrated using a single feature point at a very stable ROI of head area. They have used both the horizontal and vertical component as an input to ICA after using bandpass filtering. They have used smartphone for recording the videos.

Pros and Cons: The performance measure of this method is similar to that of the previous one. However, it is computationally efficient due to less number of feature points. Thus, this method can be used in online monitoring.

C. Statistical Mode of Analysis

The above-mentioned methods use various statistical measures to extract PR using noninvasive techniques from the raw data. Researchers have used various dimensionality reduction techniques like PCA, ICA, LE, Isomap, LLE, LTSA, or LLTSA to extract the pulse signal from a mixture of other noise component. Therefore, a brief overview about the statistical methods used in these papers is deemed to be necessary for better understanding of the PR estimation techniques.

1) Independent Component Analysis: ICA is a statistical technique used to separate the number of underlying source signals from a set of linearly dependent observed mixed signals without any prior knowledge of the source signals. Hence it is also termed as a BBS technique that separates the signal based on linearity, non-Gaussianity, and statistical independence assumptions. The ICA model assumes that the linear combination of unknown source signals is mixed in some linear fashion denoted by a mixing matrix $\mathbf{A}$ to give the observed signals $\mathbf{X}(t)$ as given in (1), where $\mathbf{A}$ is unknown

$$\mathbf{X}(t) = \mathbf{A}\mathbf{S}(t). \quad (1)$$

The aim of ICA is to find a separation or denoising matrix $\mathbf{W}$ such that it approximates the inverse of mixing matrix $\mathbf{A}$. Then, an approximate of the source signals can be estimated using

$$\hat{\mathbf{S}}(t) = \mathbf{W}\mathbf{X}(t). \quad (2)$$

Various techniques were developed for ICA [105]–[109]. However, the one that is based on the JADE [110] algorithm developed by Cardoso [106] has been found to be the most popular among all.

2) Principal Component Analysis: PCA is a typical linear dimensionality reduction technique that is most common in statistical pattern recognition and signal processing. It is a method of projecting a set of observed multivariate signals of correlated variables into some linearly uncorrelated orthogonal basis called principal components. The orthogonal basis are decided based on direction of maximum variance of the data in the feature dimensional space such that the first principle component is in the direction of maximum variance of the data points, the next component has the next largest variance maintaining orthogonality with that of the first one, and so on. The number of principle components are generally less than the number of original signals. Let $\mathbf{S}_1, \mathbf{S}_2, \dots, \mathbf{S}_p$ be the maximum variance of p linearly transformed components of N observed signals $\mathbf{X}$. Then, they can be obtained using (3), where ϕ_i are the corresponding eigenvectors of the covariance matrix $\mathbf{C} = E\{\mathbf{X}\mathbf{X}^T\}$ in accordance to first p largest eigenvalues of $\mathbf{C}$

$$\mathbf{S}_i = \phi_i^T \mathbf{X}. \quad (3)$$

3) Laplacian Eigenmap: LE [88] is a technique of unfolding the nonlinear data distribution of low dimensional data pattern in a hyperdimensional space. Owing to the nonlinear distribution of data points, they cannot be approximated to a linear basis using PCA or ICA, since linear transformation makes global transformation such as rotation, translation or scaling. Hence for preserving the local relationship between the data

points, LE employs manifold learning by solving an optimal distance minimization problem as given in (4), where y_i and y_j are two points in low dimensional space, i is among the k -NNs of j and vice-versa, and W_{ij} is the weight that measures the proximity of two point x_i and x_j in the higher dimensional space

$$\min \sum_{i,j=1}^n \|y_i - y_j\|^2 W_{ij}. \quad (4)$$

4) Other Nonlinear Techniques: The basic concept of other dimensionality reduction techniques such as Isomap, LLE, LTSA, or LLTSA is similar except the cost function equation. A generalized high-level description of these algorithms is given in Algorithm 1.

Algorithm 1: Generalized nonlinear dimensionality reduction technique.

- 1: Determining neighborhood of each points/nodes in high-dimensional space.
 - K -NNs.
 - Any distance metric for proximity neighborhood computation.
 - 2: Construction of a neighborhood graph.
 - Each node is connected to all its neighbor by an edge.
 - Edge length is given a weight.
 - 3: Compute shortest path between two nodes in a weighted graph.
 - Dijkstra's algorithm [111].
 - FloydWarshall algorithm [112].
 - 4: Computation of low dimensional embedding.
 - Multidimensional Scaling.
-

IV. EXPERIMENTAL SETUP AND DATASET

In this section, we present the description about the experimental setup, dataset, method-specific arrangements, and other parameters used in our experiments. A moderately large dataset has been prepared to benchmark the selected PR estimation techniques discussed in the preceding sections. Images and videos were recorded using a VGA web-cam attached with a laptop computer powered by a 2.2-GHz processor, 4 GB of internal memory running in an operating environment of Microsoft Windows 7 version. Videos were recorded at 30 frames/s. Video recordings and subsequent processing, e.g., RGB value extraction or head motion tracking, were done using Intel OpenCV [113] library of version 2.4.9. Signal-processing steps were implemented using MATLAB (version 2010a). The recordings of reference BVP signal were acquired using an industry-level PPG device, popularly known as Easy Pulse (version 1.1) sensor interfaced to the computer through Arduino UNO microcontroller board. A 2-min facial video was recorded for each volunteer participated during PR recording using PPG sensor. The setup that was used for such recordings are shown in Fig. 1. A flow diagram of the proposed implementation can be found in Fig. 2.

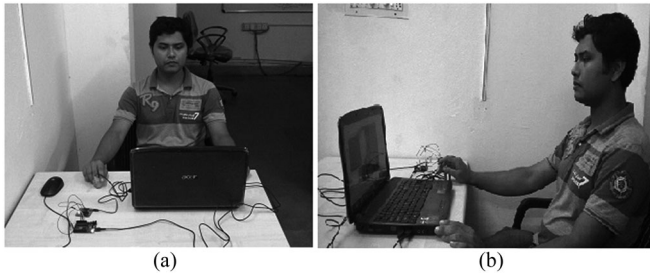


Fig. 1. Experimental setup that has been used to facilitate simultaneous recording of PPG signal and facial video.

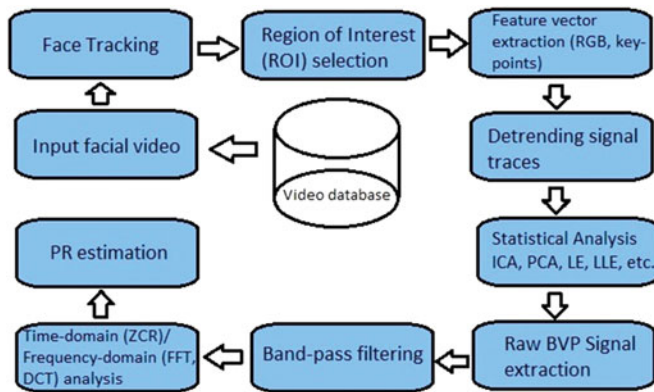


Fig. 2. Flow diagram showing a generalized methodological PR analysis technique.

The experiments were conducted with a total of 14 (four females and ten males) volunteers belonging to 22–35 years of age group. Five sets of data (each 2-min duration) were recorded for every participant corresponding to five different environmental conditions. We have named them as follows: Condition A, Condition B, Condition C, Condition D, and Condition E. For the first four conditions, the experiments were conducted in ambient lightning condition using florescent tubes normally used in lightning up laboratory complexes, while the last one was performed on daylight within an experimental hall. Out of the four artificial lighting setups, the first one corresponds to adequate florescent lightning condition with an approximate distance of 1.5 ft. between the participant and the camera. The second one was performed in a low-lightning condition keeping the distance unchanged. The third one was performed keeping the lightning condition same as Condition A; however, the distance was doubled. The fourth condition was similar to that of the first one except the participants were asked to vary their expressions during recording. The participants were asked to be sited in a relax mood keeping the head as stable as possible. We have not used any special lightning during video recordings.

A. Customization of Method-Specific Parameters

After carefully analyzing the techniques given in Table I, we have selected five existing baseline methods that cover major contributions in this field of research. These methods are

BSS-based independent component separation developed by Poh [10], PCA-based techniques developed by Lewandowska *et al.* [13], nonlinear analysis approaches proposed by Wei *et al.* [14], and two motion-based techniques developed by Balakrishnan *et al.* [15] and Irani *et al.* [16]. To prepare data for testing Poh's method [10], we selected the ROI as the center of the face. This essentially minimized the chance of nonface pixels included in the computation. A noise reduction technique was applied to remove motion artifacts from each components by subdividing the video into temporal segments. Next, estimation of PR was carried out separately for each segment. If the PR estimate obtained from the present segment was not within a range of ± 12 beats/min, measurement was taken from the previous segment ignoring the present one. If no such frequency component was found within the operational range, previous PR estimate was considered as the present PR estimate. Lewandowska *et al.* [13] have used an operating range of 0.5–3.7 Hz in their experiments; however, to keep consistency with other methods, we adjusted the range to 0.75–4 Hz. Also, we kept the ROI sufficiently large to reduce the noise component as mentioned in their paper. The implementation of Wei's method [14] was done by keeping an operating range of 0.75–4 Hz as compared to 0.7–4 Hz to make the comparisons even. While experimenting with Irani's method [16], we used a small-sized moving average window that was not considered in Balakrishnan's method [15] and we used DCT instead of FFT.

V. EXPERIMENTAL EVALUATION

A comparative analysis has been carried out to benchmark the techniques discussed in Section IV. Analysis has been done using Bland–Altman plots [114] as a combined graphical and statistical interpretation of accuracy of the measurements from the video against the reference ground truth signals. The measurement difference in between the PR estimates from the video and the sensor is plotted using the average value. The plot shows the mean of the absolute difference along with 95% limit of agreements (± 1.96 SD), where SD represents standard deviation of the difference of two measurements. The RMSE and Pearson's correlation coefficient (r) have been calculated to understand the consistency of the methods on varying environmental conditions. The overall analysis can broadly be categorized into two parts, namely Analysis-I and Analysis-II. The first category is mainly for image-analysis-based PR measurement techniques that are further categorized into linear and nonlinear methods, whereas the second type of analysis is done across various motion-based techniques.

A. Analysis-I: Image-Guided Analysis

The image-analysis-based techniques have been implemented by considering the whole face as the ROI and face and forehead portions have been marked manually. The objective behind such an analysis using two different ROI's was to study about the optimal selection of ROI's and to compare their relative performance. Lewandowska *et al.* [13], forehead region is expected to produce better results as compared to face. This is because of the homogeneity and uniform visibility of the region. Sec-

TABLE II
PERFORMANCE COMPARISON OF THE IMPLEMENTED METHODS USING FIVE DIFFERENT CONDITIONS

Conditions	Condition A		Condition B		Condition C		Condition D		Condition E		Overall	
Color-Based Methods	RMSE (beats/min)	r	RMSE (beats/min)	r	RMSE (beats/min)	r	RMSE (beats/min)	r	RMSE (beats/min)	r	RMSE (beats/min)	r
From FACE												
ICA 1 st [10]	11.15	0.26	16.49	0.11	8.72	0.48	18.67	0.18	15.71	0.18	14.52	0.02
ICA 2 nd [10]	5.36	0.85	10.36	0.61	7.41	0.62	7.27	0.82	4.82	0.87	8.15	0.67
ICA 3 rd [10]	10.25	0.42	15.1	0.35	7.93	0.59	8.36	0.49	7.08	0.74	9.63	0.56
PCA+ZCR [13]	7.28	0.61	9.11	0.5	8.22	0.8	8.77	0.7	3.69	0.92	7.67	0.72
PCA+FFT [13]	18.51	0.43	15.9	0.02	18.88	0.36	11.69	0.3	11.78	0.28	15.67	0.32
LE [14]	8.46	0.64	10.86	0.55	6.4	0.92	5.31	0.9	4.73	0.84	7.5	0.77
Isomap [14]	17.88	0.5	26.04	0.15	20.56	0.49	15.45	0.21	12.76	0.44	19.09	0.03
LLE [14]	21.65	0.09	19.45	0.33	16.15	0.07	14.42	0.51	14.42	0.53	17.46	0.18
LTSA [14]	13.71	0.29	17.39	0.08	14.59	0.38	16.4	0.44	10.51	0.45	14.71	0.33
LLTSA [14]	10.87	0.54	16.68	0.07	11.56	0.3	11.56	0.38	7.11	0.81	11.95	0.45
LDA [14]	25	0.03	18.69	0.29	16.04	0.06	22.42	0.28	21.56	0.26	20.97	0.08
From FOREHEAD												
ICA 1 st [10]	21.55	0.26	14.32	0.06	15.41	0.14	19.83	0.05	15.79	0.6	17.6	0.07
ICA 2 nd [10]	8.11	0.62	13.39	0.23	9.26	0.33	10.29	0.54	7.23	0.69	10.19	0.43
ICA 3 rd [10]	12.05	0.34	13.73	0.29	10.75	0	13.12	0.17	12.72	0.05	12.27	0.28
PCA+ZCR [13]	8.65	0.6	16.42	0.37	8.29	0.58	14.68	0.49	8.15	0.73	11.79	0.53
PCA+FFT [13]	25.93	0.68	20.6	0.14	20.78	0.23	27.14	0.05	24.22	0.24	23.88	0.22
LE [14]	6.52	0.75	8.3	0.81	5.07	0.85	4.62	0.88	1.94	0.98	5.7	0.84
Isomap [14]	18.02	0	14.58	0.49	15.93	0.09	26.73	0.18	15.02	0.09	18.61	0.04
LLE [14]	15.93	0.3	21.05	0.17	18.5	0.03	17.8	0.38	8.86	0.62	16.94	0.24
LTSA [14]	10.51	0.63	17.22	0.18	12.23	0.36	14.8	0.1	8.67	0.61	13.04	0.3
LLTSA [14]	9.11	0.74	12.3	0.54	9.68	0.52	10.89	0.47	5.6	0.84	9.78	0.57
LDA [14]	19.58	0.28	22.65	0.19	14.86	0.33	22.46	0.16	15.32	0.13	19.27	0.15
PCA+FFT [15]	10.49	0.35	12.2	0.04	10.58	0.45	13.1	0.02	9.07	0.15	11.18	0.27
PCA+DCT [16]	8.18	0.47	10.23	0.47	6.88	0.74	9.88	0.47	3.66	0.95	8.12	0.67

The table shows the accuracy and presents a comparison among various PR measurements using Conditions A, B, C, D, and E. RMSE signifies root-mean-squared error and r denotes the Pearson's correlation coefficient.

only, absence of any moving element such as lips or eyelids decreases the chance of motion artifact, resulting better PR estimation. However, a decrease in ROI's size can increase the level of noise as in most cases, the spatial average/frame from a selected ROI gets accumulated. Thus, we have considered both the ROIs in our experimentation to make a comparison as shown in Table II.

1) Evaluation Across Linear Analysis Methods:

First, we have evaluated PR obtained using all three components of ICA using face as well as forehead across five different conditions as mentioned in Section IV. It may be observed from the RMSE and correlation statistics that the second ICA produces better results in all five conditions irrespective of ROI (face or forehead) as compared to first or third components with an overall RMSE of 8.15 and 10.91 beats/min, respectively. This is consistent with the observation as reported by Poh *et al.* [10]. However, RMSE reported by Poh *et al.* (2.9 beats/min) is significantly lower than ours. Experimental setup and selection of data in their experiments might have played a decisive role. However, Poh's method actually failed on another challenging dataset, e.g., MAHANOB-HCI dataset [79], as reported by Li *et al.* [17]. However, when PCA was used instead of ICA combined with positive slope zero-crossing (ZCR) for mean PR calculation, the overall RMSE closely matched with the RMSE of ICA. This matches with Lewandowska's observation. Also there is an ambiguity regarding the selection of proper ICA component over PCA as the components

are already sorted in terms of maximum variance of data distribution in higher dimensional space. However, PR estimated from face is better than forehead in our dataset with an overall RMSE of 7.67 beats/min. It may also be observed that performance of LDA is very poor with an overall RMSE as high as 20 beats/min. In terms of correlation coefficient (r), PCA with ZCR ranks highest with 0.72 and 0.53 for face and forehead regions, respectively.

Bland–Altman analysis has been performed over all pairs of measurements across all linear techniques as depicted in Fig. 3. Significant variation between ground truths and estimated values are observed toward the mean area when face region is used as compared to forehead regions. Hence, it can be inferred that the face ROI gives a better estimate in case of these techniques. The mean difference bias is least in the case of PCA. The second independent component and PCA+ZCR have achieved best RMSE among ICA and PCA. These are highlighted in Bland–Altman plots shown in Fig. 3(a), (b), and (h).

Moreover, not only the statistical methods but also the environmental conditions play a key role. After a thorough analysis of the five conditions, we have observed that Condition E (daylight illumination) performs the best for PR estimation among all methods discussed in this paper. This is due to the fact that VGA cameras incorporate noise in the indoor lightning conditions with respect to daylight condition. Least accuracy has been observed in case of Condition B because of low-light conditions, which is expected.

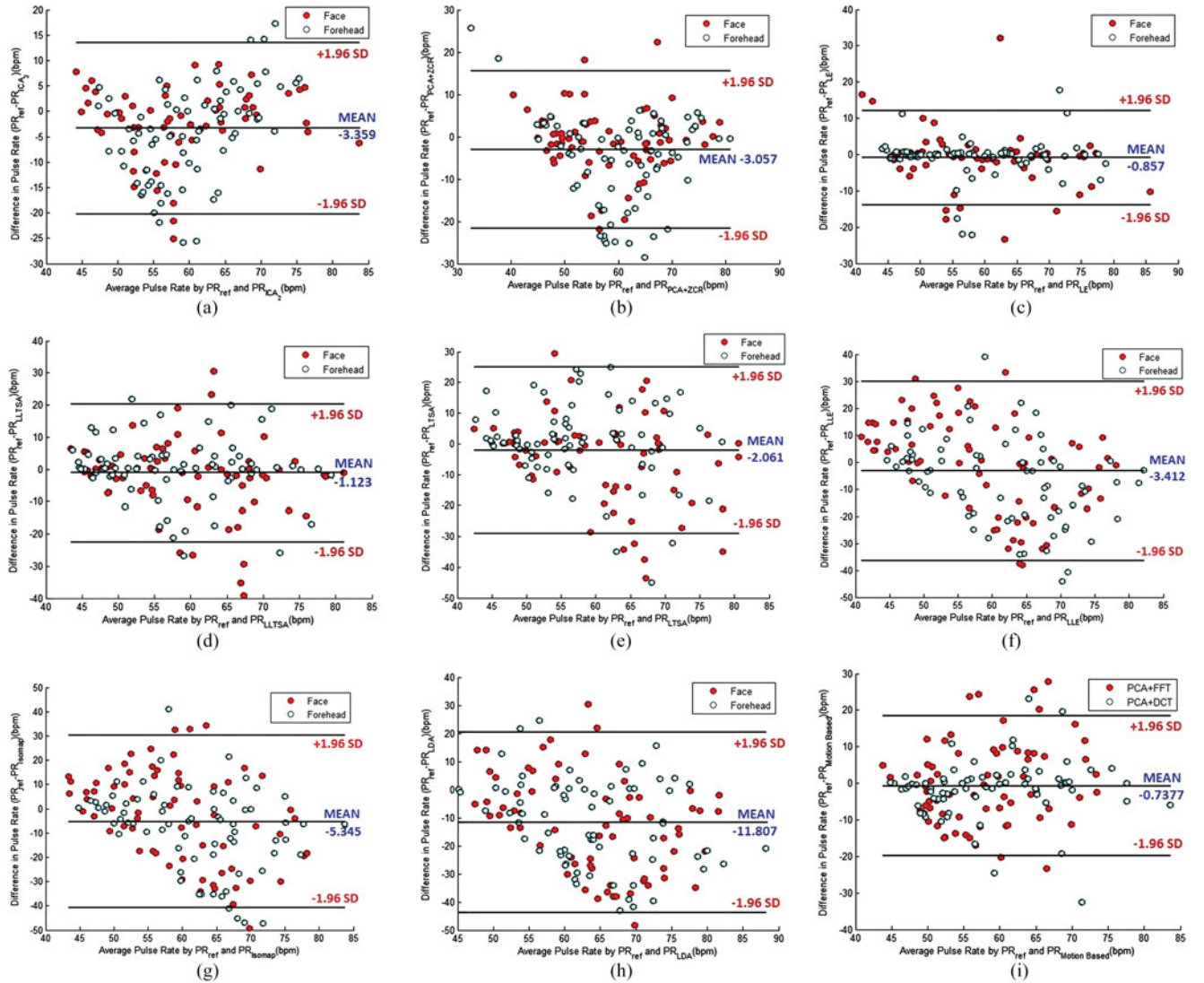


Fig. 3. Bland–Altman plots demonstrating the degree of agreement between the PR measured from video and PPG sensor. Plots (a)–(h) show overall comparison between errors of face and forehead regions using different color-based methods, whereas plot (i) shows the overall comparison of motion-based techniques. The lines correspondingly represent the upper limit of agreement (+1.96 SD: standard deviation), mean, and lower limit of agreement (−1.96 SD: standard deviation)

2) Evaluation Across Nonlinear Analysis Methods:

We have implemented the nonlinear techniques based PR estimation as mentioned by Wei *et al.* [14]. Primary objective of using nonlinear techniques for pulse estimation is to unfold nonlinearly warped PR data in a high dimensional space. The following nonlinear methods, namely LE, LLE, LTSA, LLTSA, and Isomap, have been implemented. It has been observed that as per the RMSE statistics, LE performs significantly better as compared to all linear as well as nonlinear techniques with an overall RMSE of 4.73 beats/min (face) and 1.94 beats/min (forehead) as mentioned in Table II. LLTSA [14] came close to linear methods with RMSE vary between 6 and 12 beats/min. Among all the linear and nonlinear techniques, LE have attained the highest correlation coefficient r of 0.98 and least RMSE of 1.94 beats/min when Condition E (daylight) is used.

Bland–Altman plots (c)–(g) of Fig. 3 shows the pictorial representation nonlinear techniques for PR measurements. It can

be easily verified from the plots that LE (over forehead) is the best with mean difference bias of −0.857 beats/min. LTSA and LLTSA also perform closely to LE; however, error remains high.

B. Analysis-II: Motion-Guided Analysis

In case of motion-based techniques, key feature points are tracked and their traces are considered for estimating the PR of the participants. These methods include the Balakrishnan *et al.*'s [15] approach which is the first one in this category that uses displacement of selected feature points in the y -direction over time using Lucas Kanade tracker [98] and the Irani's method [16] which is similar to Balakrishnan's method except it has replaced FFT by DCT and they applied a moving average filter before the Butterworth filtering. We have rejected the most erratic trajectories based on a rejection percentage α before performing PCA. Hence, there is no significance to consider the

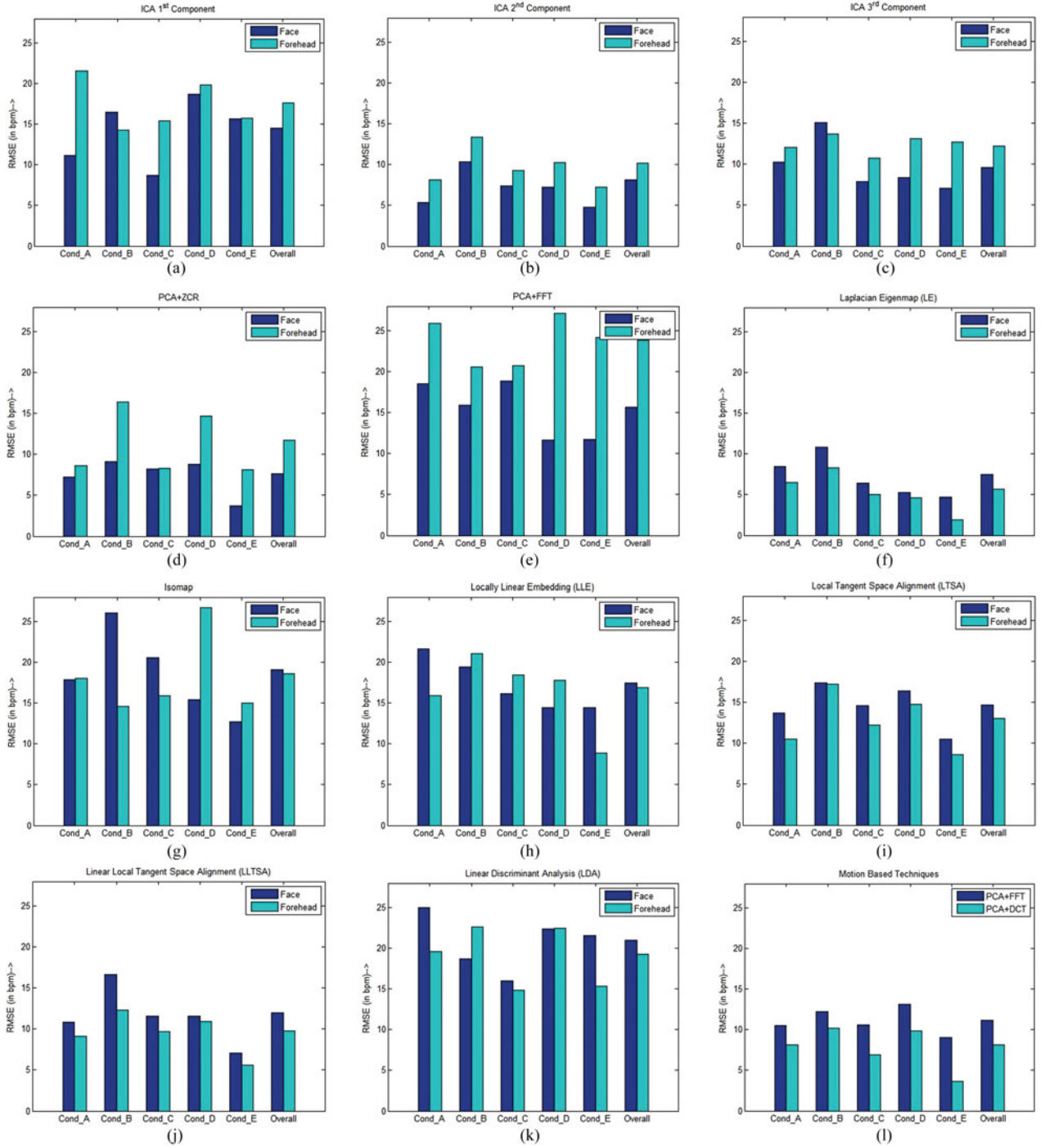


Fig. 4. RMSE plots showing comparisons of errors between PR values measured from face and forehead regions over varying conditions across different methods.

forehead region separately as both face and forehead regions are expected to produce similar motion patterns.

It can be verified from Table II that the overall RMSE score for DCT improves by 3 beats/min, while the lowest recorded RMSE is 3.66 beats/min at Condition E, which is encouraging in the context of real life applications of noncontact-based PR estimation. The last plot of Fig. 3 shows a comparison between motion-based techniques. It is observed that the DCT-based

method is more accurate than the FFT-based method when combined with PCA.

C. Analysis of Computational Complexity

Though time taken to estimate PR does not affect much on the diagnosis process, however, for instantaneous or online estimation, we prefer a method that is computationally efficient as well

as accurate. Patients with highly fluctuating PR in ambulatory conditions, i.e., newborn infants prone to epileptic seizures due to premature birth and patients admitted in critical care units, need special attention. Thus, online, noninvasive, and contactless PR estimation can be a big advantage. Thus, we must select the technique that is fast and accurate. The computational overhead of each of these techniques has been recorded during processing of a 2-min video. It is observed that the linear techniques consume negligible amount of time (< 100 ms) with PCA running in the order of 3–5 ms and ICA runs with an order of 50–80 ms. However, nonlinear techniques are computationally expensive and runs in terms of seconds that vary from 5 (LE) to 24 s (LLTSA). Hence, among the nonlinear techniques, LE ranks the best in terms of time complexity. On the other hand, the motion-based technique is faster with average times of 462 (FFT) and 82 ms (DCT). This is because the larger the number of feature vectors fed to PR extraction, more the time.

D. Overall Comparison

A pictorial representation of the accuracy of all methods is presented in Fig. 4. Here, the x -axis of the plot represents five environmental conditions for RMSE estimation using face and forehead. It can be inferred from the plots that face performs better than forehead for majority of the conditions in linear analysis, whereas forehead performs better when nonlinear techniques are used. Out of all the combinations, the LE-based technique has been found to be most promising for PR extraction, especially when the forehead region is selected as ROI. The RMSE plot of LE is also in agreement with the Bland–Altman plot. Another notable observation is that all methods perform better in daylight conditions. Fluorescent lamps give a distorted output and incorporate noise that makes the signal more stochastic than periodic. Our overall impression is that Wei's [14] method using LE outperforms all other methods. Hence, it is the best choice for unfolding the true pulse from the distorted and noisy components.

VI. DISCUSSION OF RESULTS

In this section, we present discussion on each of the methods implemented and analyzed in this paper. We acknowledge that Poh *et al.* [10], [11] were the first to propose an approach for the possibility of measuring BVP through computer-vision-assisted, contactless, and noninvasive method. They exploited the concept of PPG and modified the light source using a visible light camera. Some improvements of their method have been proposed by various research groups, such as the method proposed by Kwon *et al.* [12].

In 2011, Lewandowska *et al.* [13] have shown that if PCA is used instead of ICA, computational overhead can be reduced. They have also argued that if forehead segment is used instead of face, accuracy can be further improved. When we have verified the accuracy of their approach, we found that ICA(2) and PCA perform equally well, though face gives better estimation as compared to forehead segment. The nonlinear method using LE proposed by Wei *et al.* [14] though performs much better than other linear (PCA, ICA, LDA) and nonlinear

decomposition (Isomap, LLE, LTSA, LLTSA) techniques; however, it takes considerable amount of time when compared against linear methods.

On the other hand, the motion-based technique proposed by Balakrishnan *et al.* [15] assumes that if the maximum frequency and its second harmonic contain maximum spectral power within the operating range, the signal can be considered as the most periodic and it can be used for estimating the PR. Irani *et al.* [16] have modified Balakrishnan's method by applying a smoothing filter before the Butterworth filter and replaced FFT by DCT to handle motion artifacts. We have observed that such steps improve the results in practice. In addition, the eighth-order Butterworth filter used by Irani *et al.* may have helped to reduce the fuzziness in the boundary frequency selection.

VII. CONCLUSION

A comprehensive review on image- and video-based PR estimation methods is presented in this paper. We have implemented five state-of-the-art techniques on a homogeneous platform and evaluated them with a large dataset by varying various environmental conditions. Our study covers major technical contributions of the domain. It has been observed that none of the existing techniques can predict accurate PR at all types of conditions and, thus, may not be suitable for clinical purposes unless improved further. Out of the five techniques that have been implemented and benchmarked in this paper, LE by Wei *et al.* [14] was the most robust that is followed by the improved Balakrishnan's method (PCA+FFT) proposed by Irani *et al.* (PCA+DTC).

We believe that contributions made in this domain within this short span of time (2007–2015) is significant and it may open up new research dimensions. We also believe that with rapid improvement in optical sensor technology, researchers can extract more accurate information from skin color or head tracking. Though the techniques need further improvement before making products acceptable to the medical community, however, home-based nonclinical applications can be tried immediately.

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